NeuroLens

Watch a brain watch your video.

PREDICTED BRAIN RESPONSE TO VIDEO, SECOND BY SECOND

CS5224 CLOUD COMPUTING · FINAL PRESENTATION · DYLAN CHAN · XIN WEI CHONG · KELVIN LOW · JOSHUA LUM · BRANDON TAN

$10k+

to test one ad on a brain scanner

fMRI study	**$10 – 15k**	2–4 weeks
NeuroLens	**$1.00**	≈ 7 min

Dozens of cuts.
No way to judge them.

Generative tools made production cheap. The feed made every second count.



What already exists: measured, filmed or guessed.

Physical fMRI study university and research labs	$10,000 – $15,000	2–4 weeks
Facial coding Realeyes, iMotions webcam studies	$850 – $4,000	3–5 days
AI attention maps Neurons AI, Attention Insight	$500 – $1,500 / mo	sub-minute
NeuroLens predicted brain response	$1.00	≈ 7 min

OUR DIFFERENCE

A brain response, not a gaze or a face

Per second, with a no-sound pass

Compare any two cuts directly

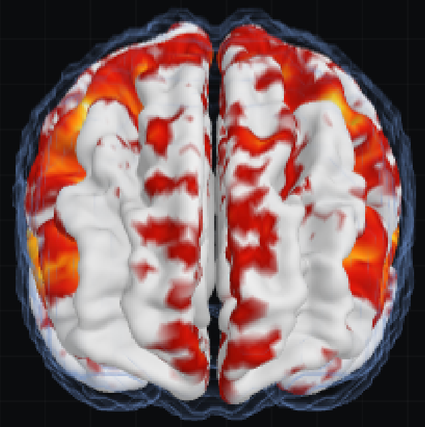
Under the hood: TRIBE v2.

Frames
V-JEPA 2 watches

Sound
Wav2Vec-BERT listens

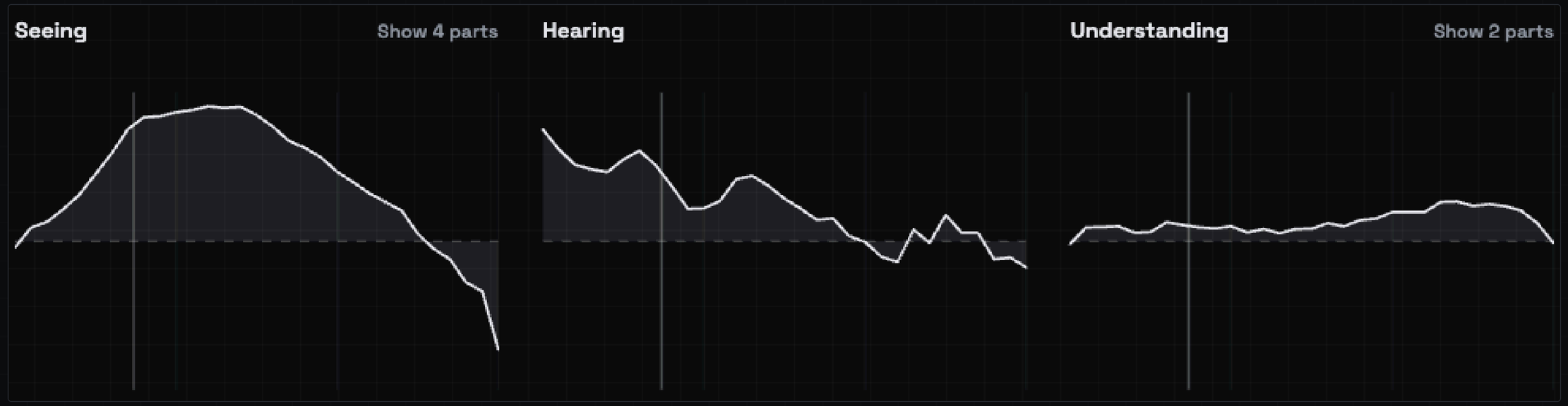
Words
WhisperX + LLaMA 3.2 read

TRIBE v2
1,000+ hours of fMRI
from 720 people



20,484 cortex points, every second

Seven signals, three questions.



SEEING

faces · bodies · scenes · visual

HEARING

sound

UNDERSTANDING

speech · social

BELOW TYPICAL

0 = TYPICAL FILM & TV VIEWING

ABOVE TYPICAL

Pay as you go. Credit, not cards.

$1.00

per started 30 s of video

NEW USERS: $10 CREDIT = TEN 30 S VIDEOS

WHY PREPAID CREDIT: CARD FEE AS A SHARE OF THE CHARGE

A $1.00 video, paid by card

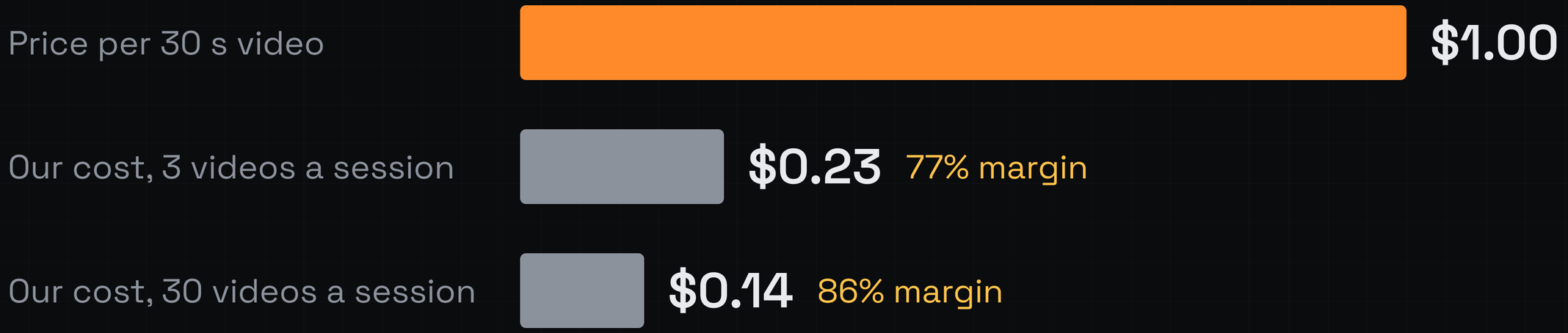
≈ 33%

A $20 credit pack

≈ 4%

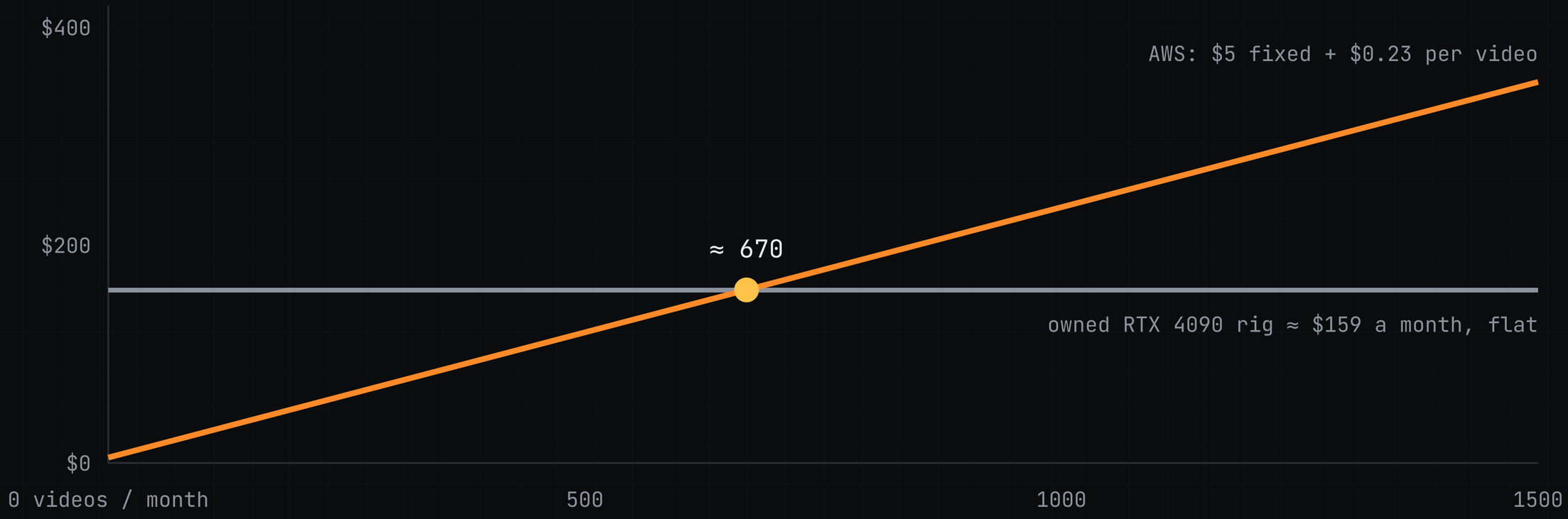
COURSE VERSION: AN ADMINISTRATOR GRANTS CREDIT · STRIPE CHECKOUT BELONGS TO THE PRODUCTION DESIGN

What one analysis costs us.

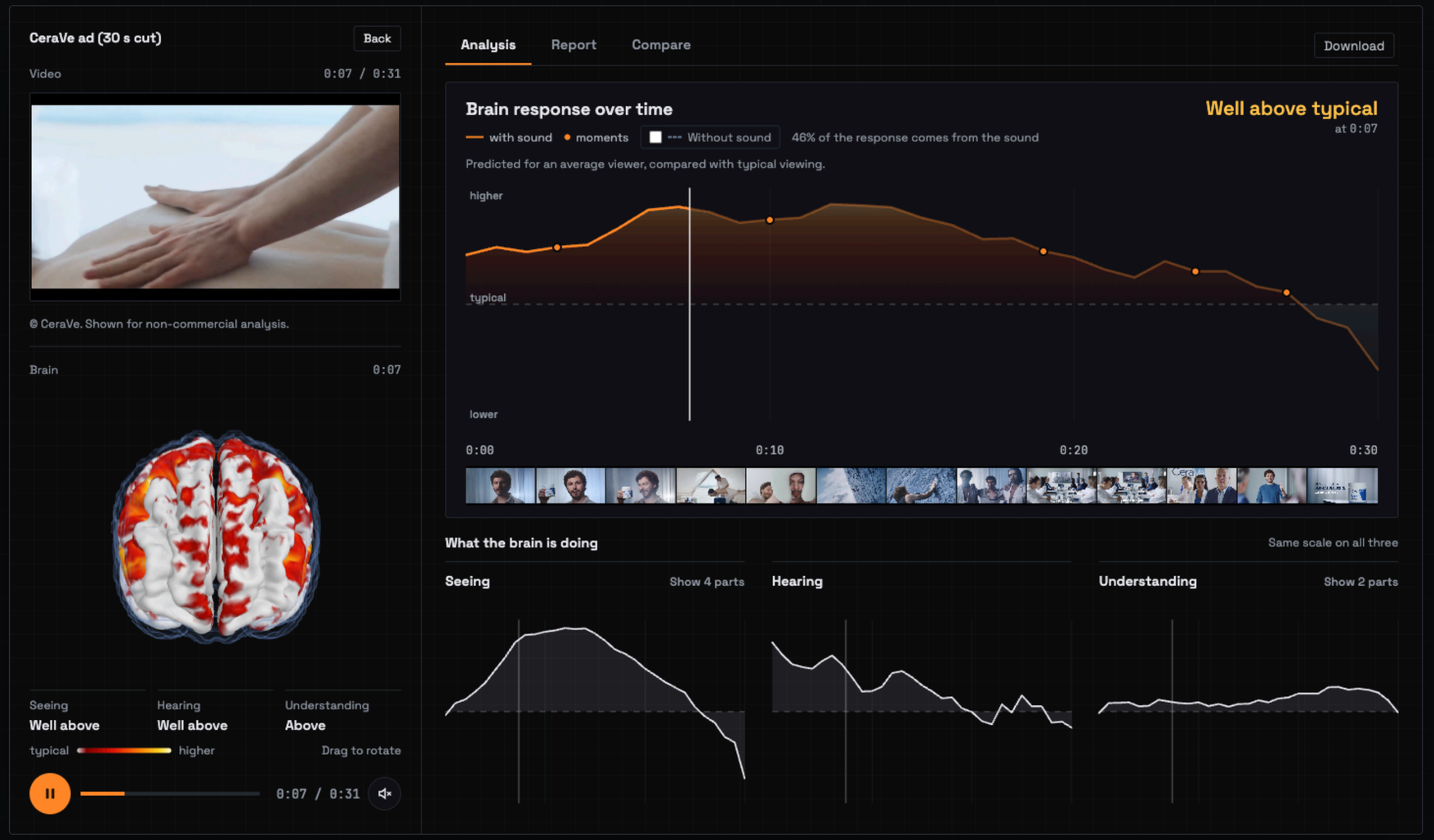


GPU 393 S AT $1.22/H (L4) · START-UP AND IDLE SHARED ACROSS A SESSION · FIXED COST $5 A MONTH

Cloud is cheaper below about 670 videos a month.



LIVE Live demo: sign in, upload, explore



1. Upload: price first
2. Peak, dip, no-sound toggle
3. Seeing · hearing · understanding
4. AI report, compare, bad file

CLICK THE PICTURE TO OPEN THE LIVE SITE · DF6GIRQYYKYFA.CLOUDFRONT.NET

Can we trust it?

SAME ANSWER ON TWO MACHINES

0.005

Largest difference across 11 series, same clip on our RTX 4090 and on AWS. Every series correlates at 1.000.

FAILED JOBS ARE REFUNDED

77 / 80

Automated tests: a failed or killed job refunded exactly once. The other 3 stopped on a GPU-library error in setup.

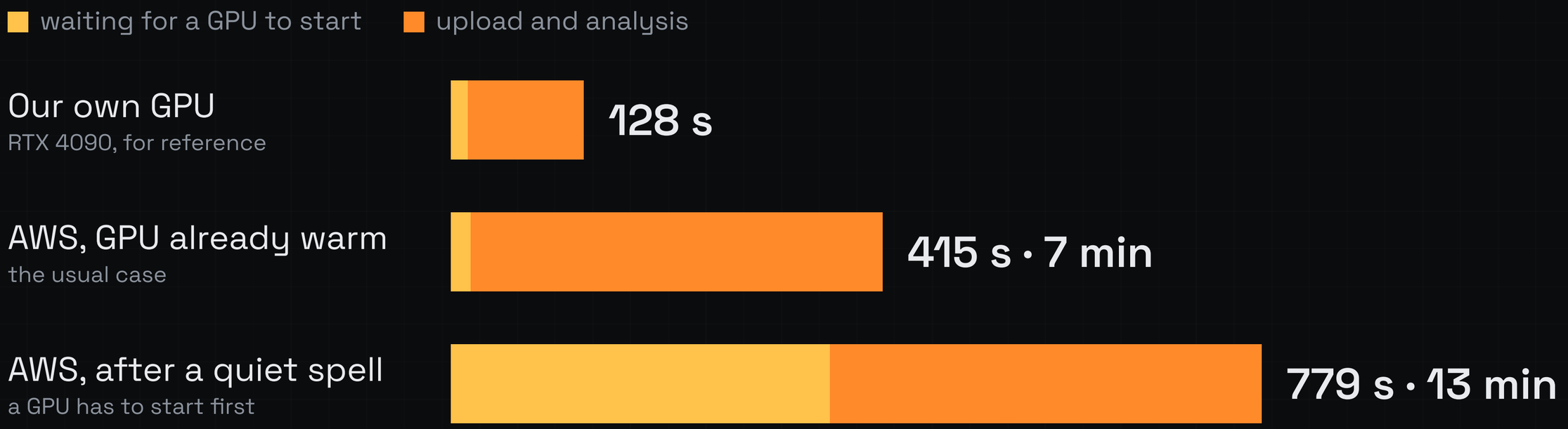
NOT TESTED

Views, sales

Whether a higher response means more views or sales. We make no such claim.

TRIBE V2'S ACCURACY IS META'S RESULT (NEW VIDEOS, TASKS AND PEOPLE); WE DID NOT RE-TEST IT AGAINST REAL SCANS

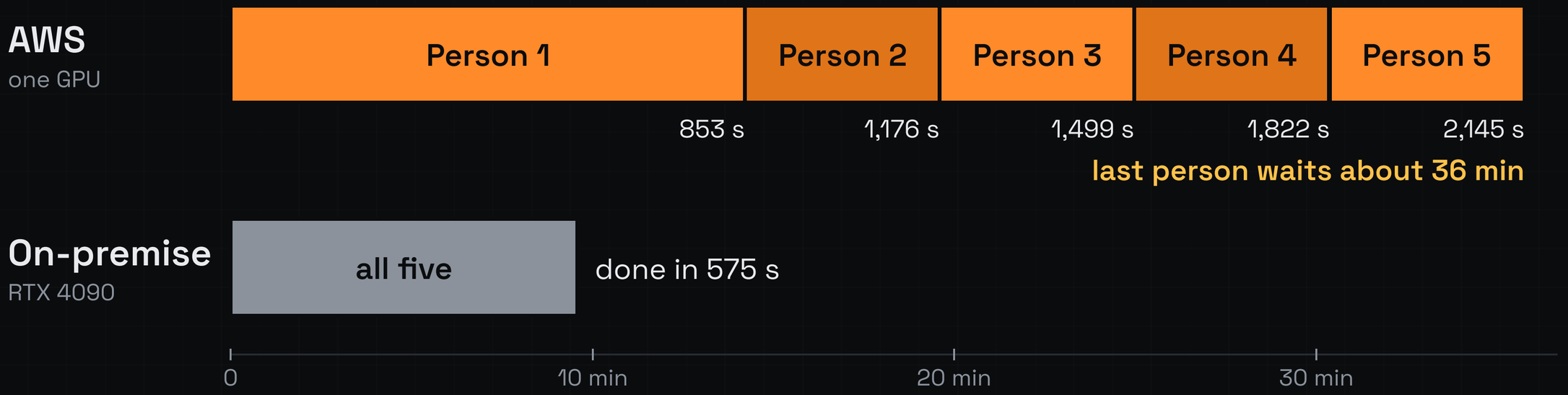
How long will a user wait?



SAME 30 S CLIP · GPU WORK PER VIDEO-SECOND: 3.7 S ON THE RTX 4090, 10.7 S ON AN A10G · COLD-START WAITS: 452 S AND 277 S

Five people at once: they wait in line.

Five uploads at the same moment. One GPU runs one job at a time.



NOTHING WAS DROPPED. THE AWS RUN GOT AN A10G, AND ITS FIRST JOB APPEARS TO HAVE RESTARTED ONCE.

What broke, and what we did.

FIXED

Words repeated in every chunk

Before: all words in every chunk



After: each chunk, its own words

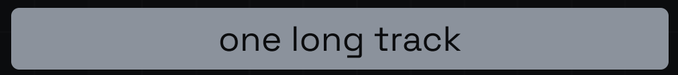


We replaced TRIBE v2's speech step.

FIXED

Long audio ran out of GPU memory

Example: 150 s of audio



Split into equal pieces ≤ 60 s

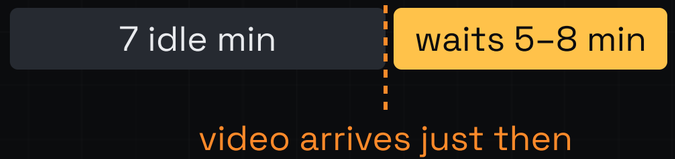


We split audio over 60 s into equal pieces.

KNOWN LIMIT

A video arrives as the GPU stops

The GPU stops after 7 idle min



The request restarts the GPU, so it waits a further 5–8 min.

Can people use it?

Booth visitors tried a sample or uploaded a clip, then answered the 10-question SUS.

[TBD] / 100

SUS SCORE · 68 IS AVERAGE · 80 IS VERY GOOD

[X]% SAW SOMETHING THEY HAD NOT NOTICED
[Y]% WOULD PAY $1.00 FOR A 30 S VIDEO

What it claims, and what it doesn’t.

WHAT IT IS

- An average viewer’s brain response
- One scale for every video

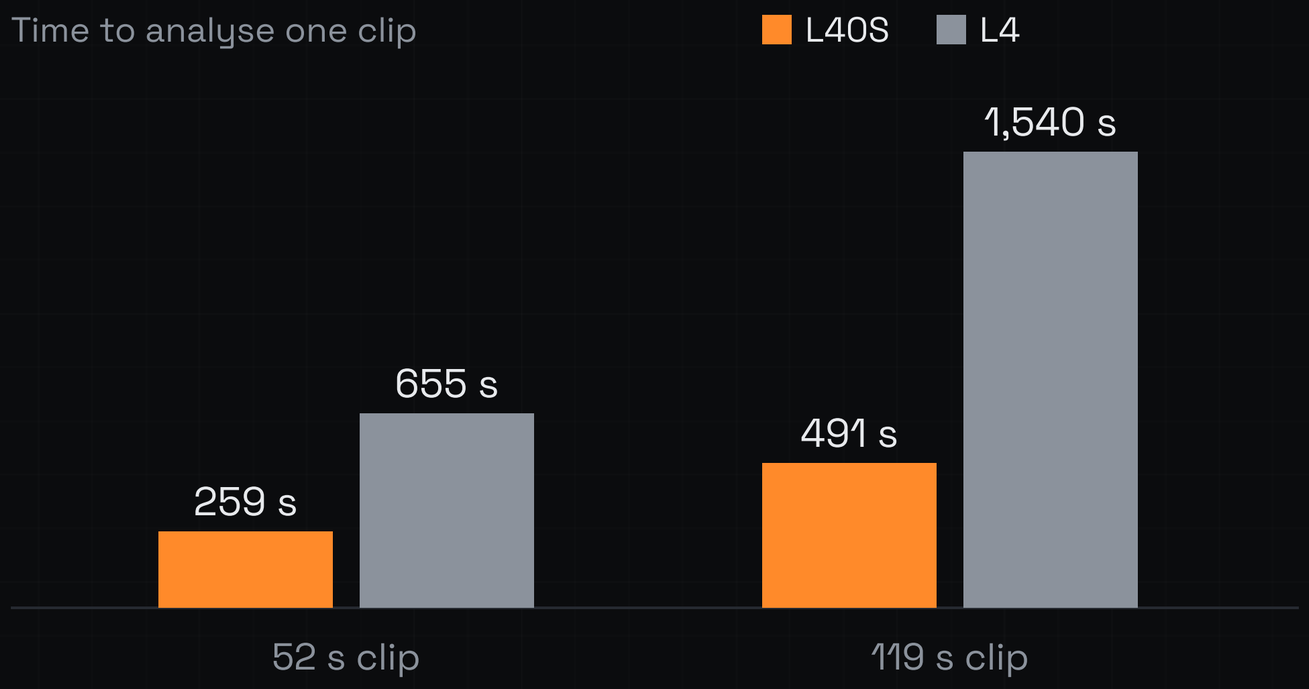
WHAT IT ISN'T

- A prediction of views or sales
- Instant: first video ≈ 6 min

ETHICS

- Non-commercial licence (CC BY-NC)
- No one scanned or profiled
- Anonymous, consented survey

Where next: scaling up.



26 / 26

Spot launches failed for lack of capacity

19.9 GB

peak GPU memory: a 24 GB card fits

An EC2 GPU pool: $0.10 – $0.20 a video, about $70 more a month. It pays off above about 1,000 videos a month. Estimates, not built.

A $10,000 brain study, in 7 minutes for $1.

$1.00
per 30 s video

≈ 7 min
to a result, GPU warm

670
videos a month: break-even vs owning

≈ 6 min
first video after a quiet spell

Next

Licence

CC BY-NC: a paid launch needs one.

Next

Scale

A GPU pool that grows with the queue.

Next

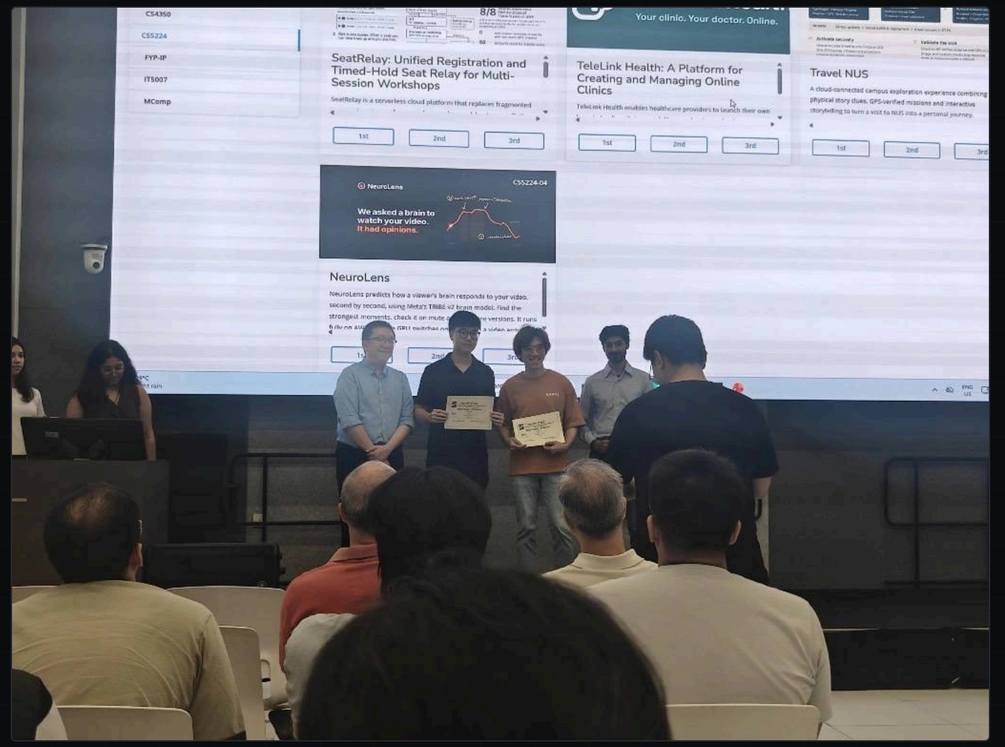
Polish

Fix what our usability study flagged.

Thank you. Questions?

df6girqyykyfa.cloudfront.net

CODE · GITHUB.COM/KELVINLOWHM86/NEUROLENS



Thanks for supporting us at STePS!